

# THIN CERAMIC FERROELECTRIC PHASE SHIFTER FOR STEERABLE MICROSTRIP PATCH ANTENNA ARRAY

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## ABSTRACT

The possibility of having an electronically steerable antenna system is of significant current importance for many communication technologies. Applications such as airport traffic control, satellite tracking for mobile communication system, and radar systems emphasize the importance of electronically steerable antennas. A novel design using ferroelectric material which dramatically reduces the size and the cost of the phase shifter used to achieve an electronically steerable antenna system is presented in this paper. A total beam scan of  $36^\circ$  is obtained for a two element microstrip antenna system while the side lobe level is kept below 10 dB, and the reflection coefficient at resonance frequency is maintained below 20 dB.

## 1. INTRODUCTION

Due to the high cost and large dimensions of low frequency phase shifters, the use of an electronically steerable antenna system operating at low frequencies has been limited. This paper proposes the use of thin ceramic ferroelectric materials for the design of phase shifters used in the realization of the electronically scanned antenna system, as a means of overcoming the limitations of current technology. A simple and novel design

employing two ferroelectric phase shifters in conjunction with two microstrip antennas operating at 2.1 GHz is presented, and measurement results of reflection coefficient and radiation pattern are provided in this paper. The design goal was to obtain as much as possible beam scan, while minimizing the amplitude of the sides lobes and the reflection coefficient ( $S_{11}$ ). Because for this prototype example only two microstrip elements are used, a maximum scan of  $36^\circ$  ( $18^\circ$  on each side) can be achieved, if a side lobe level below 10 dB is desired. Increasing the total number of antenna elements, allows for larger scan for the same side lobe level. The use of ferroelectric material in the microwave frequency range has been limited in the past due to high losses of these materials and due to the high electric field necessary to bias the structure in order to obtain substantial dielectric constant change. We demonstrated in previous work how a significant reduction in material losses is possible by using a new sol-gel technique [1, 2] to produce Barium modified Strontium Titanium Oxide ( $\text{Ba}_{1-x}\text{Sr}_x\text{TiO}_3$ ) which has ferroelectric properties at room temperature. We also demonstrated how the use of thin ceramics reduces the required bias voltage, with almost no power consumption required

to induce a change in the dielectric constant [3, 4].

## 2. SYSTEM DESCRIPTION

Fig. 1 shows the layout of the system including the two microstrip patch antennas printed on RT-DUROID having relative dielectric constant of 2.33 and thickness of 1.5 mm. The two disks are the phase shifters made of Barium modified Strontium Titanium Oxide thin ceramic, their detailed layout is reported in Fig. 1. As observed the two extremes of the microstrip line printed on the ceramic disks have patches which facilitate the connection of the phase shifter with the rest of the circuit using wire bonds. To reduce the parasitic capacitance associated with the larger patches, the ground plane of the sample extends only under the strip as visible in Fig. 1. DC-blocks are used to insulate the static bias field from the RF feeding circuit. The DC-block and their corresponding connecting arms were designed to obtain a good match with  $50\ \Omega$  system. For the presented configuration the two patch antennas will operate under static bias field. This can be avoided if desired with the use of two additional DC-blocks placed after the phase shift before the antennas. The static bias to change the ferroelectric material dielectric constant is provided with two high impedance transmission lines having length which is multiple of a quarter wavelength in the substrate. This guarantee RF insulation from the static circuitry. Because ferroelectric materials have loss which decreases under bias conditions [5], it is convenient to keep the material under bias when no phase shifter is desired (hard bias) while reducing

the voltage (soft bias) to obtain the phase shift.

## 3. MEASUREMENT RESULTS

Measurements of return loss and radiation pattern for the system described in Fig. 1 were conducted in free space. Fig. 2 shows the results of the measured reflection coefficient, for soft biased ( $V_b=250V$ ) and hard biased ( $V_b=450V$ ) of the BST phase shifter used in the array, while the antenna beam is scanned. Good matching is obtained for the array system in both conditions at resonant frequency, with a minimum reflection coefficient below 20 dB.

Measured result for the power radiation pattern under three different bias conditions is reported in Fig. 3. We observe a total phase tilt of the main beam of  $18^\circ$  on each side, corresponding to  $36^\circ$  of total scan. Even further scan can be achieved by biasing the phase shifter harder [4]. This particular design, is not desired in order to keep the second main beam level below 10 dB.

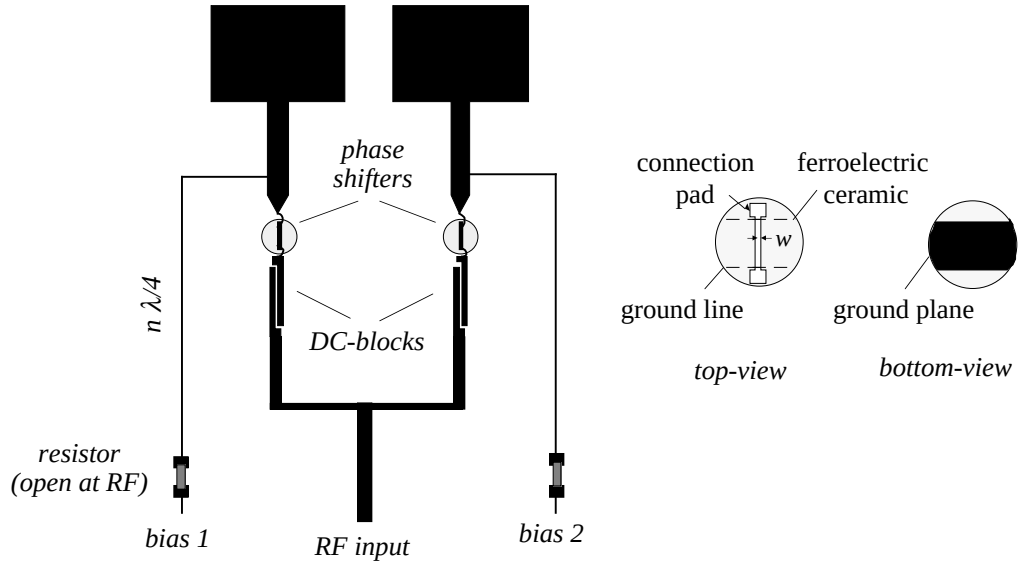
Also the bias voltage for thin ceramic is relatively high (up to 450V in this design), compared with standard IC circuits, since the current required to maintain or rapidly shift the phase is very low, the overall control power requirements (about 4 mW) are at least one order of magnitude less than those for comparable ferrite phase shifters. This is due to the fact that in ferroelectric materials the driven energy required to change the properties of the material goes primarily to the change in the stored electrostatic energy and is not dissipated in the ferroelectric material.

#### 4. CONCLUSIONS

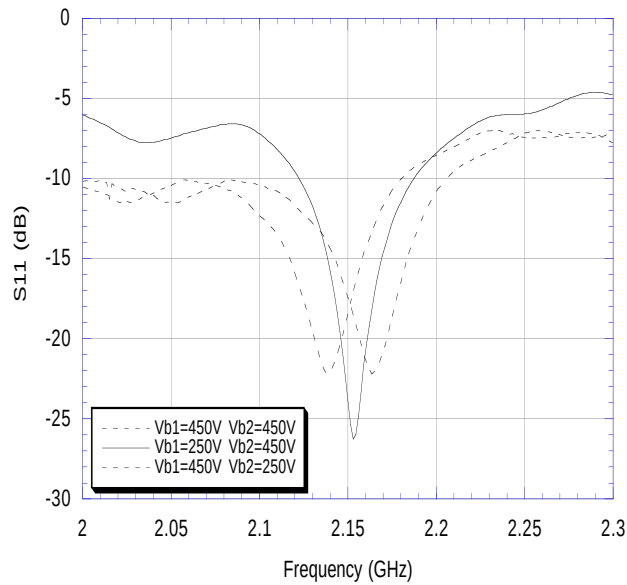
A novel type of phase shifter based on thin ceramic ferroelectric material is integrated in a planar array microstrip patch antenna system and the overall performance of the array is evaluated. The measurement results of reflection coefficient and radiation power, shows that this novel type of phase shifter can efficiently be used in planar microstrip antenna circuits. A total beam scan of  $36^\circ$  was achieved using a two element array system, while maintaining a reflection coefficient below 20 dB, and a side lobe level below 10 dB.

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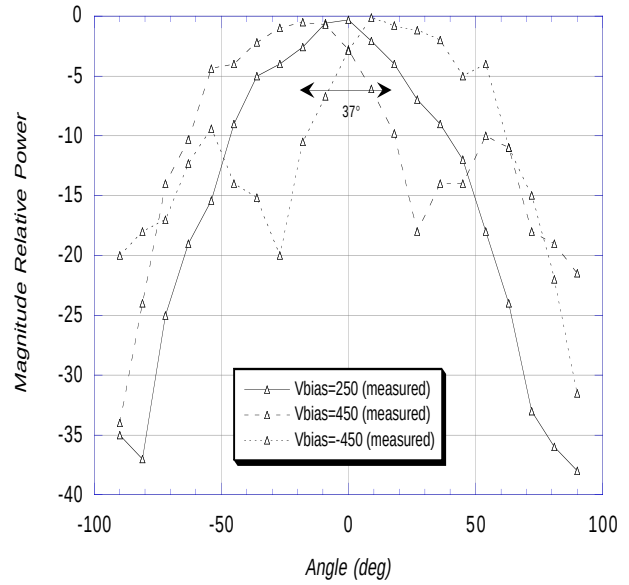
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**Fig.1** Layout of the two element microstrip array antenna system



**Fig. 2** S11 measurements for the array under soft and hard polarization



**Fig. 3** Measured radiation pattern for the array system under different bias condition